Clàudia Janó Muñoz

PhD student - University of Cambridge

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keywords: Transit photometry; Astronomical detectors; Late-type stars

Research interests

I use high-precision, ground-based near-infrared photometry to find and characterise planets around the coolest stars, working across the whole chain from detector to demographics. During my PhD I optimised the calibration and reduction pipeline of SPIRIT, an InGaAs CMOS detector at SPECULOOS-South, and characterised its long-term on-sky performance; this work underpins the validation of new planets such as TOI-2407 b, a warm Neptune in the Neptune desert. I apply the same photometry to stellar and substellar variability, and I lead a collaboration combining MeerKAT radio and SPECULOOS infrared monitoring of brown-dwarf magnetospheres. Building on earlier machine-learning work on photoevaporation and the radius valley, I aim to connect instrument-limited precision to population questions, in particular the Neptune desert.

Employment and Education

2026 – 2027	**Postdoctoral Researcher**, ETH Zurich.
2022 – 2027	**PhD in Physics**, Cavendish Laboratory, University of Cambridge. Thesis: *Optimisation, performance and first discoveries of the SPIRIT infrared detector for exoplanets transiting late-type stars.* Supervised by Prof. Didier Queloz and Dr. Matthew Hooton. Funded by the Harding Distinguished Postgraduate Scholarship. Submission: February 2027.
2018 – 2022	**MPhys Physics with Astrophysics**, First Class, University of Manchester.

Research experience

Oct. 2022 – present	**PhD research**, Cavendish Laboratory, University of Cambridge (SPECULOOS consortium). • Optimised the calibration pipeline and characterised the on-sky performance of SPIRIT, a novel InGaAs CMOS near-infrared detector on the Callisto telescope (SPECULOOS-South). Modelling and correcting for dark current and glow, row-pattern effects, bad pixels, noise, and operating temperature trade-off. • Analysed the first year of SPIRIT operations of ultracool dwarfs and TESS candidates (first-author, RASTI, under review). • Validated the Neptune desert planet TOI-2407 b with multi-band ground-based follow-up including SPIRIT (first-author, MNRAS 2025). • Co-developing `goibniu`, a machine learning-based observation tool to optimise the scheduling of observations with SPIRIT by mitigating the effect of bad pixels and maximise the number of comparison stars (Hooton & Janó Muñoz, in prep()).
Oct. 2021 – May 2022	**Master's project**, University of Manchester. Machine learning in spectral analysis of ALMA observations. Supervised by Prof. Gary Fuller.
Sep. 2021 – Oct. 2021	**Summer research project**, Universitat Autònoma de Barcelona. Simulating the effects of close stellar encounters with N-body simulators. Supervised by Dr. Ignasi Ribas.
Jun. 2021 – Aug. 2021	**UROP Summer Internship**, Imperial College London. Exoplanet atmospheric evolution with machine learning (led to Rogers, Janó Muñoz et al. 2023, MNRAS). Supervised by Dr. James Owen.

Grants and awards

October 2022	**Harding Distinguished Postgraduate Scholarship** (£20,000 per year of study), including a £2,000 yearly stipend for academic travel.
December 2018	**Physics Success Scholarship** (£2,000), for academic excellence in physics and mathematics.

Observing and proposals

July 2026	**Co-I**, *Unlocking Ultracool Dwarf Magnetospheres via Simultaneous Radio/IR Monitoring*, MeerKAT radio telescope, jointly with CfA doctoral student Kevin Ortiz-Ceballos (outcome pending).
Oct. 2022 – Jan. 2025	Regular remote observing with the Callisto and Ganymede telescopes, SPECULOOS Southern Observatory.
Dec. 2023	On-site observations, instrument maintenance and testing at the SPECULOOS Southern Observatory, Cerro Paranal, Chile.

Services

Committees and organisation

Jan. 2025 – Apr. 2025	**International Women's Day Committee**, Member, Institute of Astronomy, University of Cambridge.
Sep. 2023 – Sep. 2025	**Queloz Group**, Organiser and host of the group's meetings, events, and journal club, Cavendish Laboratory, University of Cambridge.

Outreach

Mar. 2025	**International Women's Day**, Ran an exoplanets workshop for high-school students, Institute of Astronomy, Cambridge.
Mar. 2025	**Open Day**, Volunteer, Institute of Astronomy, Cambridge.

Teaching responsibilities

Jun. 2026 – Sep. 2026	**Teaching Assistant**, Cambridge Centre for International Research, UK.
Apr. 2024 – May 2024	**Part III Example Classes: Exoplanets Course**, Cavendish Laboratory, University of Cambridge.
Nov. 2022 – May 2023	**Part IA Physics Supervisions**, Emmanuel College, University of Cambridge.

Talks

Seminars

March 2026	*SPIRIT: A CMOS Detector for Exoplanets Around Ultracool Dwarfs*, Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA, USA.
November 2025	*SPIRIT: A CMOS Detector for Exoplanets Around Ultracool Dwarfs*, Exoplanet Seminar, Observatoire de Paris-PSL, France.
August 2022	*Emulating photoevaporation with Machine Learning*, University of California Santa Cruz, USA (remote).

Conference talks

February 2026	*Search for habitable exoplanets with the SPIRIT infrared detector*, Harding Scholar Conference, Cambridge, UK.
January 2026	*First Results from SPIRIT: A CMOS Detector for Exoplanets around Ultracool Dwarfs*, Rocky Worlds IV, Groningen, Netherlands.

Posters

June 2026	*Ground-based transits in the infrared with SPIRIT*, Exoplanets VI, Porto, Portugal.
June 2024	*Ground-based transits in the infrared with SPIRIT*, Exoplanets V, Leiden, Netherlands.
January 2024	*Finding transiting planets in the infrared with SPIRIT*, Rocky Worlds III, ETH Zurich, Switzerland.

Departmental and consortium talks

June 2026	*Brown dwarf radio synergies*, SPECULOOS consortium meeting, University of Liege, Belgium.
December 2025	*Discussion on SPIRIT: Infrared CMOS Photometry for Exoplanets around Ultracool Dwarfs*, LCLU discussion series, Cambridge, UK.
November 2025	*Discovery of the Neptune-like planet TOI-2407 b*, Harding Scholars Internal Speaker Event, St Catharine's College, Cambridge, UK.
February 2025	*Exoplanet Detection with SPIRIT: Infrared CMOS Photometry and the Discovery of the Hot Neptune TOI-2407 b*, Hills Coffee Talks, Battcock Centre for Experimental Astrophysics, Cambridge, UK.
February 2025	*TOI-2407 b: A hot Neptune in the Neptune Desert discovered with the SPIRIT Infrared Detector*, Exoplanet Seminar series, University of Cambridge, UK.

Software development

`SPECULOOS pipeline`	Co-developer: Data reduction pipeline for the SPECULOOS consortium, including optimised SPIRIT data reduction and analysis.
goibniu	Co-developer: machine-learning-based observation tool for SPECULOOS, under development (Hooton & Janó Muñoz, in prep.).
SMART	Co-author: machine-learning tool for fitting protostellar spectra.

Extracurricular activities

May 2021 – Feb. 2022	**Astronomy Society**, Chair, University of Manchester.
Feb. 2020 – May 2021	**Astronomy Society**, Science Officer, University of Manchester.

Skills

Programming	**Proficient**: Python; **Intermediate**: C++
Computing	**Experienced**: LaTeX, Linux, Docker; **Basic**: HPC
Languages	**Native**: Catalan, Spanish; **Proficient**: English; **Intermediate**: German (B1)

Publications

10 refereed or accepted journal papers (1 first-author, 1 second-author), 1 first-author paper under review, 2 conference proceedings.

First and second author

under review	**Janó Muñoz, C.**, et al. *Performance and prospects of the SPIRIT infrared CMOS detector for exoplanet detection*, RASTI
June 2025	**Janó Muñoz, C.**, et al. *TOI-2407 b: a warm Neptune in the desert*, Monthly Notices of the Royal Astronomical Society, 541, 630–641
March 2023	Rogers, J. G., **Janó Muñoz, C.**, Owen, J. E., Makinen, T. L. *Exoplanet atmosphere evolution: emulation with neural networks*, Monthly Notices of the Royal Astronomical Society, 519, 6028–6043

Contributing author

September 2026	Scott, M. G., et al. *TOI-6981 b: a transiting sub-Neptune at the water ice line*, accepted to Monthly Notices of the Royal Astronomical Society, arXiv:2609.37597
September 2026	Lang, F. Z., et al. *TOI-4616 b: an Earth-sized planet transiting a nearby M4 dwarf*, Monthly Notices of the Royal Astronomical Society
July 2026	Pedersen, P. P., et al. *Astra: an open-source fully autonomous robotic observatory control software*, arXiv e-prints, arXiv:2607.12898
April 2026	Dransfield, G., et al. *MANGOS – II. Five new giant planets orbiting low-mass stars*, Monthly Notices of the Royal Astronomical Society, 547, stag448
March 2026	Scott, M. G., et al. *Two temperate Earth- and Neptune-sized planets orbiting fully convective M dwarfs*, Monthly Notices of the Royal Astronomical Society, 547, stag070
December 2025	Barkaoui, K., et al. *TOI-7166 b: a habitable zone mini-Neptune planet around a nearby low-mass star*, Monthly Notices of the Royal Astronomical Society, 544, 2637
October 2025	Zúñiga-Fernández, S., et al. *Two warm Earth-sized exoplanets and an Earth-sized candidate in the M5V-M6V binary system TOI-2267*, Astronomy & Astrophysics, 702, A85
July 2024	Gillon, M., et al. *Detection of an Earth-sized exoplanet orbiting the nearby ultracool dwarf star SPECULOOS-3*, Nature Astronomy, 8, 865–878
October 2023	Triaud, A. H. M. J., et al. *An M dwarf accompanied by a close-in giant orbiter with SPECULOOS*, Monthly Notices of the Royal Astronomical Society: Letters, 525, L98–L104

Conference proceedings

August 2024	Zúñiga-Fernández, S., et al. *SPECULOOS: five years hunting terrestrial planets around ultra-cool dwarfs*, Proceedings of the SPIE, 13094
July 2024	Pedersen, P. P., et al. *Infrared photometry with InGaAs detectors: first light with SPECULOOS*, Proceedings of the SPIE, 13096